

NeuroTrace-DAG

When next-item metrics mislead: a multi-dataset evaluation study in educational logs

FINAL PROJECT SCOPE	Multi-dataset evaluation-validity and protocol study
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Executive summary

This project audits whether next-item recommendation metrics in educational logs reflect educational progression or simply the repeat behavior generated by a platform. It introduces a reproducible protocol that separates continue, revisit, and advance actions; compares simple baselines; uses leakage controls and sequence-level confidence intervals; and keeps negligible and unresolved findings in the final report.

What we found	Evidence-based conclusion
Action structure	Concept-level repetition differs strongly across datasets, so aggregate item metrics are not directly comparable.
Prototype ranking	RAGR - Recent-5 Recall@5 = 0.0067 [0.0045, 0.0095]; below the practical threshold 0.01.
DAG readiness	Delta AUC = 0.0005 [0.0002, 0.0008]; status: detectable_but_negligible.
Forgetting	Gap>=1-day delta AUC = 0.0003 [-0.0011, 0.0016]; status: unresolved.

The contribution is the audit protocol, not a claim that the prototype is a superior teaching recommender.

Dataset and implementation flow

The study uses three public educational logs after aligning the comparison to a concept unit: Junyi topics, ASSISTments skills, and XES3G5M knowledge components (KCs). This prevents raw exercise or question identifiers from creating a misleading cross-dataset comparison.

Dataset / unit	Transitions	Continue	Revisit	Advance
Junyi topic	2809159	93.0%	4.0%	3.0%
ASSISTments skill	214725	67.9%	18.4%	13.6%
XES3G5M KC (method A: question→primary KC)	3975344	26.2%	31.3%	42.5%
XES3G5M KC (method B: drop multi-KC extras)	3975344	26.2%	31.3%	42.5%
XES3G5M KC (method B parquet is_repeat=0)	4410668	26.3%	30.2%	43.5%

Actual data flow

Public log and expert graph -> sequence-level split -> action relabelling (continue / revisit / advance) -> baseline and prototype ranking -> sequence-level paired bootstrap -> practical-threshold interpretation -> final audit JSON -> this report.

Leakage safeguards

The evaluation keeps a fixed sequence sample, records duplicate-sequence checks, computes confidence intervals by resampling sequences rather than individual rows, and distinguishes what can and cannot be tested when learner identifiers are unavailable.

How the DAG is used

The directed acyclic graph encodes declared prerequisite relations. It is evaluated as a readiness feature for correctness prediction and visualized as a learning-map structure. Observational correlation is not presented as proof that forcing the graph improves learning outcomes.

Results and interpretation

Analysis	Measured output	Interpretation
Ranking	RAGR R@5 0.891; Recent-5 0.885	The difference is statistically detectable but below the predeclared practical bar.
Readiness	Delta AUC 0.0005; practical bar 0.005	Detectable but negligible; do not make a causal pedagogy claim.
Forgetting	Success-only AUC 0.661; +log gap AUC 0.661	Gap information has unresolved value in the tested data.

What the live interface shows

The interface is an exploratory prototype viewer. A user enters a short practice history, the system returns a scored continue/revisit or advance action type, ranks candidate concepts, and shows nearby prerequisite relationships. It is intentionally labelled as model inspection rather than a validated lesson prescription.

Accurate method labels

The local neural rank probes are not faithful reproductions of published GRU4Rec or SASRec. They are retained only as GRU-sequence and attention-sequence exploratory probes. This avoids overstating a small two-model experiment as a benchmark comparison.

Viva-ready explanation

We found that next-item accuracy can reward a platform's repetition structure rather than meaningful educational progression. Our contribution is a reproducible evaluation protocol that makes this visible across topic, skill, and KC units, compares simple recency baselines, and reports practically negligible or unresolved results honestly.

Research integrity and future work

Claims deliberately excluded

- state-of-the-art recommender
- RAGR practically outperforms simple recency baselines
- the current probe is a faithful GRU4Rec or SASRec reproduction
- DAG readiness provides a practically meaningful correctness gain
- forgetting features improve prediction on the tested data

What would strengthen this into a teaching-intervention study

A future study should preregister a learner-level intervention, compare the recommendation policy against robust baselines, and measure downstream learning gain rather than only the next logged click. That evidence is required before claiming the system improves student learning.

Key references

Ren et al. (2019), RepeatNet: A Repeat Aware Neural Recommendation Machine.

Repeat/explore gating predates RAGR; cite it instead of claiming the gate as novel.

Hidasi et al. (2016), Session-based Recommendations with Recurrent Neural Networks.

Original GRU4Rec uses a session-recommendation ranking protocol; the local two-epoch probe is not a reproduction.

Kang and McAuley (2018), Self-Attentive Sequential Recommendation.

Original SASRec is a separate validated method; the local attention probe is not a published SASRec baseline.

Reproducibility

Source of truth: `outputs_junyi/phases/research_integrity_audit.json`. Regenerate with `scripts/build_research_integrity_audit.py` followed by `scripts/render_research_integrity_docs.py` and this PDF builder.